

Paid Events Backend

Phase 1 Backend Scope & Pricing Proposal — Paid Events & Payment Engine

- **Target Delivery:** 9 working days from acceptance and environment access
- **Scope:** Production-Ready Backend Engine, Database Schemas, Webhooks, Admin Actions, and Feature Test Suite

1. RECONCILIATION WITH PRIOR PROPOSALS (JUNE & JULY ESTIMATES, S-5 SCREEN)

To ensure 100% transparency, every item from prior discussions and packages is accounted for below:

- **July Backend Estimate Reconciliation:** The July Backend estimate was **\$560.00** (18 – 24 hours) covering Event Booking, Card Hold, and Price A/B Testing. Price A/B testing has been removed (-\$120.00 / 4 – 6 hours). The remaining net reused base is **\$440.00** (16 – 20 hours).
- **June Proposal Reconciliation:** Reuses the `PaymentCaptured` and `PaymentRefunded` domain event concepts proposed in our June offer at **zero additional charge** for event definition. Only the outbound signed HTTP dispatch, queue retries, and admin replay UI are quoted as new work in Item 3.
- **S-5 Screen Reconciliation:** The S-5 (Subscription & Payment) screen from the original package was not delivered (UI scope retained by Client team per agreement). Nothing from S-5 is reused or double-charged here; this proposal reflects exclusively new backend engine capabilities.

Component / Task	Description	Included / Excluded	Estimated Hours	Amount (\$)
Stripe PaymentIntent Hold Authorization	Setup Stripe Checkout in payment mode (<code>mode => 'payment'</code> , <code>payment_intent_data.capture_method = 'manual'</code>).	Included: Gateway session creation, card capture collection. Excluded: UI/Blade styling.	6 – 8 h	\$175.00
Event Booking & Participant Lifecycle	Parent-Child Event linking (<code>parent_event_id =</code> Pool/Announcement, Child = Confirmed Crew), participant join validation.	Included: Event participant state machine (<code>pending</code> , <code>confirmed</code>).	5 – 7 h	\$150.00

Component / Task	Description	Included / Excluded	Estimated Hours	Amount (\$)
		Excluded: Frontend drawer modals.		
Stripe Base Webhook Integration	Webhook endpoint, Stripe-Signature verification header handling, basic payload routing.	Included: CSRF exemption, signature security. Excluded: Credit system transactions.	4 – 6 h	\$115.00
SUBTOTAL (Reused Base)	Net Reused Base after removing A/B Testing (-\$120.00)		16 – 20 h	\$440.00

2. ITEMIZED NEW BACKEND REQUIREMENTS (PHASE 1)

All new requirements requested for Phase 1 are detailed below with explicit inclusions and exclusions.

Item / Requirement	Technical Description	Included / Excluded	Estimated Hours	Amount (\$)
Item 1: Outing Payments & Action Audit History	Dedicated <code>outing_payments</code> table (separated from credit <code>transactions</code>), <code>payment_action_logs</code> append-only audit trail.	Included: <code>amount_cents</code> integer precision, <code>tax_profile_key</code> for myDATA classification, immutable price snapshot per signup, price edit lock on outings with signups, persisting Stripe's actual <code>capture_before</code> timestamp (not hardcoded), Card states (<code>pending</code> , <code>authorized</code> , <code>captured</code> , <code>released</code> , <code>expired</code> , <code>failed</code>), IRIS states (<code>pending</code> , <code>paid</code> , <code>refund_pending</code> , <code>refunded</code> , <code>rejected</code>), audit log recording <code>admin_user_id</code> , <code>action</code> , <code>reason</code> , <code>payload</code> , <code>ip</code> , <code>timestamp</code> , documented read contract (Query DTOs) for client visibility pages, and independent payment-attendance read query (no automatic	5 – 7 h	\$140.00

Item / Requirement	Technical Description	Included / Excluded	Estimated Hours	Amount (\$)
		financial trigger from attendance). Excluded: Modifying existing <code>iris_payments</code> or <code>transactions</code> tables, frontend UI.		
Item 2: Partial Capture Failure & Retry Engine	Per-person idempotent capture execution (<code>capture: {outing_id}:{user_id}</code>), <code>needs_attention</code> crew state transition.	Included: Filament Admin actions for Retry Payment (individual retry) and Replace Member (drop failed member & capture backup candidate from pool), manual fallback crew cancellation. Excluded: AI automated matching algorithm, frontend modal styling.	4 – 6 h	\$120.00
Item 3a: myDATA Signed Tax Webhook (BASIC)	Synchronous outbound HTTP POST webhook dispatched on <code>capture</code> and <code>refund</code> .	Included: Reuses <code>PaymentCaptured</code> / <code>PaymentRefunded</code> June event concepts (\$0 charge for event definitions); outbound HTTP POST payload with HMAC SHA256 header signature, payload format per spec. Excluded: Queued retries, database delivery logs, admin retry UI. <i>(Risk: Remote receiver outages result in lost receipt notifications).</i>	3 – 4 h	\$80.00
Item 3b: myDATA Signed Tax Webhook (DURABLE - RECOMMENDED)	Queued background delivery via Laravel Queue (<code>SendTaxReceiptWebhookJob</code>), exponential backoff retries (5 attempts over 2 hours).	Included: Reuses <code>PaymentCaptured</code> / <code>PaymentRefunded</code> June event concepts (\$0 charge for event definitions); <code>tax_webhook_deliveries</code> database log table, Filament Admin table widget displaying undelivered messages	6 – 8 h	\$180.00

Item / Requirement	Technical Description	Included / Excluded	Estimated Hours	Amount (\$)
		with manual Replay / Retry Webhook action. Excluded: Direct connection to Greek AADE/myDATA servers (handled on client side), VAT/tax calculations.		
Item 4: Dynamic Payment Settings & Provider Toggles	Integration with existing AppSetting model to store payment defaults, active provider toggles, and runtime enablement validation.	Included: AppSetting schema wiring, active payment method validation on checkout initialization, default fallbacks for new outings. Excluded: Frontend settings page UI layout (handled by Client team per agreement).	2 – 3 h	\$60.00

3. TOTAL FIXED PRICE & DELIVERY SUMMARY

Option	Included Components	Total Estimated Hours	Fixed Price (\$)	Delivery Timeline
Phase 1 — Option A (With Basic Tax Webhook)	Reused Base (\$440) + Item 1 (\$140) + Item 2 (\$120) + Item 3a (\$80) + Item 4 (\$60)	30 – 40 h	\$840.00	7 Working Days
Phase 1 — Option B (With Durable Tax Webhook - RECOMMENDED)	Reused Base (\$440) + Item 1 (\$140) + Item 2 (\$120) + Item 3b (\$180) + Item 4 (\$60)	33 – 44 h	\$940.00	9 Working Days

4. ARCHITECTURAL REQUIREMENTS, ACCEPTANCE CRITERIA & WARRANTY

Architectural Requirements (Included in Phase 1 Engine Design):

- 1. Isolated Provider Modules & Common Interface:** Each payment provider (Stripe, Viva/IRIS) lives as an isolated adapter module behind a common contract (**PaymentServiceInterface**) with declared capabilities (**authorize** , **capture** , **release** , **refund**).
- 2. Dynamic Admin Toggles:** Admin ability to enable or disable individual payment methods dynamically without code deployment.

3. **Database Settings Integration:** Utilizing the existing `AppSetting` model to store payment defaults and configuration (no duplicate settings system created).
4. **Uniform Payment Recording:** Uniform database logging on `outing_payments` regardless of provider, with `provider` stored as an explicit field.
5. **Per-Outing Payment Mode:** Storing `payment_mode` per outing (defaulting to hold-then-capture).
6. **Interface Documentation:** Delivery of a concise architecture document detailing the payment contract and adapter patterns for future provider extension.
7. **Read Contract & Shared Attendance Visibility:** Exposing documented read-only query methods for client UI pages; payment and attendance data are queryable together per participant at the read layer with no automatic financial actions triggered by attendance status.

Acceptance Criteria Covered in Engine & Feature Test Suite (`tests/Feature/OutingPaymentTest.php`):

1. **Double Click Protection:** DB-level row locks (`lockForUpdate()`) and idempotency keys prevent duplicate card charges on rapid button clicks.
2. **Duplicate / Out-of-Order Webhooks:** Idempotency gate (`wasAlreadyProcessed`) and state machine checks prevent duplicate execution or out-of-order state corruption.
3. **Expired Hold Handling:** Checking Stripe `capture_before` dates prior to capture attempts; clean status transition to `expired`.
4. **Partial Capture Failure:** Execution isolation per participant, `needs_attention` state transition, individual retry and replace-member admin actions.
5. **Refund Replay Guard:** `refund:{payment_id}` idempotency key prevents double refunds.
6. **Manual IRIS Refund:** Filament admin action to mark IRIS payments as `refunded`, creating an append-only audit log entry and firing the refund tax webhook.
7. **Permissions & Audit History:** Admin policy checks (`is_admin`) and immutable `payment_action_logs` recording admin ID, action type, reason, payload, IP, and timestamp.
8. **Simultaneous Outings Isolation:** Strict scoping by `event_id` preventing price, currency, or payment state leakage between concurrent outings.
9. **Price Freeze Enforcement:** Price changes blocked on outings after the first signup exists; capture execution strictly uses the immutable price snapshot stored at signup.
10. **Fake Participant Exclusion:** Synthetic or fake participants (`fake_participant_count`) are strictly excluded from payment charges, hold captures, and crew financial calculations.

Warranty Terms:

- **30 Days from Production Acceptance:** Includes free, priority correction of any bug or deviation from the 7 Architectural Requirements and 10 Acceptance Criteria listed above.
- **Scope Boundary:** Warranty strictly covers deviations from the agreed criteria and architecture. New feature requests or scope additions outside this specification are quoted separately.

5. DEPENDENCIES, ASSUMPTIONS & OUT-OF-SCOPE BOUNDARIES

To ensure a smooth delivery and clear operational boundaries, the delivery timeline and fixed pricing rest on the following standard engineering terms:

Delivery Dependencies & Schedule Adjustments:

- **Timeline Clock:** The 9 working day delivery window begins upon formal proposal acceptance and agreement on the milestone schedule.
- **Client Testing & Feedback Window:** Timeline assumes client review and acceptance testing feedback within 3 business days of staging deployment.
- **Timeline Pauses:** The delivery clock pauses for external blocking events beyond developer control (e.g., staging server unavailability, delayed client feedback, third-party payment gateway outages, or unforeseen health emergencies / force majeure).

Explicitly Out of Scope (Client Side / Excluded):

1. **Frontend UI & Blade Templating:** All user-facing Blade views, HTML/CSS styling, AlpineJS modal/drawer animations, and responsive UI layouts (handled by Client team).
2. **Infrastructure & Server Provisioning:** Server setup, DevOps, SSL certificate management, domain DNS configuration, or environment provisioning.
3. **Production Deployment Execution:** Direct production deployment operations or third-party merchant account setup (Stripe Dashboard settings, Viva Merchant portal self-care setup).
4. **Greek Tax Authority Integration:** Direct myDATA / AADE tax portal API communication, VAT calculation, or credit-note invoicing (handled by Client's existing receipt infrastructure).
5. **Manual Data Backfills:** Manual database migrations of legacy test payments or historical participant records.

6. OPTIONAL ITEMS (INDICATIVE NON-BINDING ESTIMATES FOR PHASE 2)

These items are non-binding estimates for future planning and are NOT included in the Phase 1 fixed price.

Item / Feature	Technical Summary & Dependencies	Indicative Estimated Hours	Indicative Range (\$)
1. Long-Window Payment Mode (Excursions 1-	Immediate card charge upon signup with an automated refund policy window, or delayed hold opening close to the date. Depends on Phase 1 engine.	12 – 16 h	\$300.00 - \$400.00

Item / Feature	Technical Summary & Dependencies	Indicative Estimated Hours	Indicative Range (\$)
2 Months Ahead)			
2. IRIS Automation with Viva	Direct Viva API integration for automated bank transfer verification instead of manual screenshot review. Depends on Viva API access and merchant setup.	14 – 18 h	\$350.00 - \$450.00
3. Refund as Credit Towards Next Outing	Ledger option to issue event credit balances instead of cash refunds, avoiding gateway processing fee losses.	6 – 8 h	\$150.00 - \$200.00
4. PayPal Gateway Adapter	Adding a PayPal SDK payment adapter behind the common payment interface. Demonstrates modular multi-provider extensibility.	10 – 14 h	\$250.00 - \$350.00